

Software PS:

Problem Statement 1

Unified AI Agri-Vision Platform for Crop Advisory and Livestock Management

Marginal farmers often lack access to specialized veterinary and agricultural expertise through a single unified platform. Existing solutions are fragmented, requiring farmers to use separate systems for crop management and livestock monitoring, resulting in lower adoption and operational inefficiencies. Develop a digital platform that enables farmers to access crop disease identification, livestock monitoring, historical farm records, and actionable advisory services through an intuitive and accessible interface suitable for rural environments.

Problem Statement 2

Smart Fleet Coordination and Logistics Management Platform

Transportation and logistics networks often suffer from inefficient vehicle utilization, poor route planning, lack of real-time visibility, and fragmented coordination among stakeholders. These inefficiencies lead to increased operational costs, delays, and resource wastage. Develop a software platform that enables real-time fleet monitoring, route optimization, load allocation, and shipment tracking to improve operational efficiency and resource utilization across logistics networks.

Note: Consider building better optimized systems with little compromise to accuracy

Hardware PS:

Problem Statement 3

Intelligent Need-Based Pesticide Sprinkler System

Conventional pesticide spraying methods often apply chemicals uniformly across entire fields, regardless of the actual health condition of individual plants. This results in excessive pesticide usage, increased operational costs, potential environmental damage, and long-term degradation of soil and water quality. There is a need for an intelligent and targeted spraying system that can identify plant health conditions and apply pesticides only where required.

Teams are required to design and develop a smart pesticide spraying solution capable of assessing plant health and selectively dispensing pesticides based on the detected level of infection, disease, or pest infestation. The solution should aim to minimize chemical wastage, improve spraying efficiency, and promote sustainable agricultural practices.

Problem Statement 4

Real-Time Renewable Energy Microgrid Monitoring System

Small-scale renewable energy installations such as solar-powered microgrids play an important role in providing reliable electricity to rural and remote communities. However, these systems often lack continuous monitoring mechanisms, making it difficult to detect performance degradation, battery failures, abnormal power consumption, or reduced generation efficiency. As a result, faults frequently go unnoticed until they significantly impact power availability and system reliability.

Teams are required to design and develop a real-time monitoring solution for renewable energy microgrids capable of tracking power generation, energy consumption, and overall system performance. The solution should provide timely insights into operational health and support the identification of anomalies, inefficiencies, or potential failures, enabling proactive maintenance and improved reliability of renewable energy infrastructure.

Bonus tip: Try to build some notification services on smartphones etc. for Hardware problems.